Weather and taxi demand in New York City: Part 1 (Spark SQL)

DISS-DPA Fall 2026. Muhammad Umar Azhar and Wadan Ali. **GitHub repository:** https://github.com/umariscoding/dpa-taxi-weather

1. Overarching question and Part 1 sub-question

Overarching question (whole project): which New York taxi zones are the most important, and does bad weather change taxi demand there more than past data predicts?

Part 1 sub-question: how do rain, snow and temperature change trips, trip time and fares per zone?

2. Data

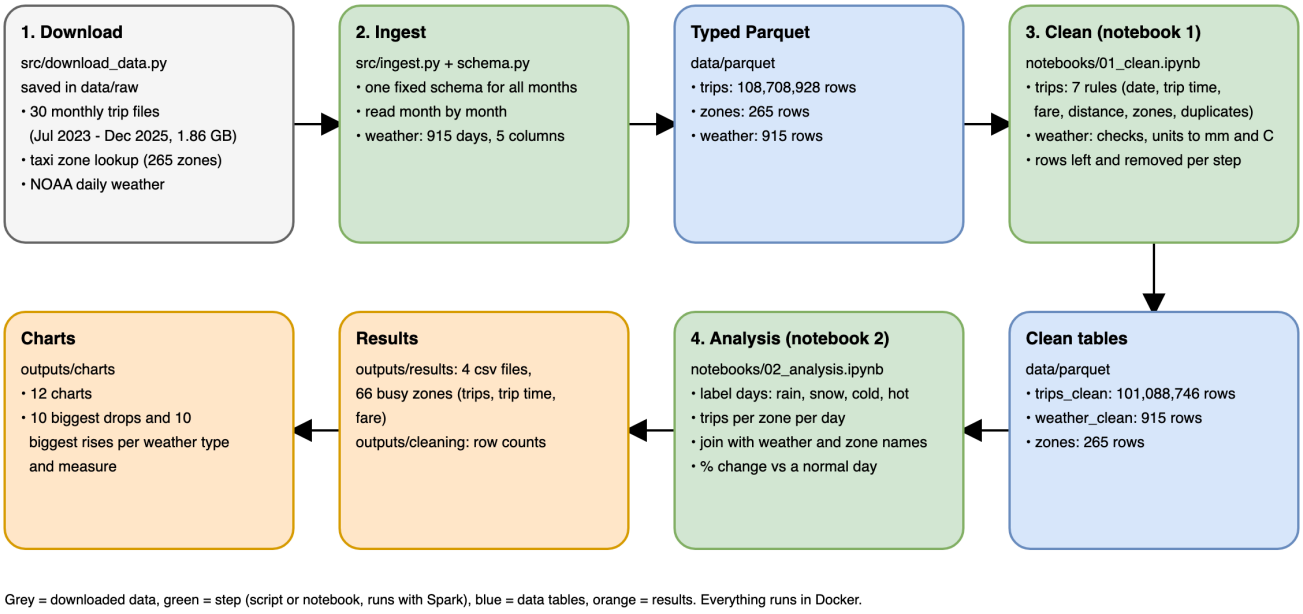
Dataset	Content	Size	Licence
NYC yellow taxi trips (TLC)	30 monthly Parquet files, Jul 2023 to Dec 2025, 108,708,928 trips	1.86 GB (all raw files)	NYC Open Data terms of use
Taxi zone lookup (TLC)	265 zones with borough and name	12 KB	NYC Open Data terms of use
NOAA daily weather, Central Park (station USW00094728)	rain, snow, max and min temperature, we use 915 days (2023-07-01 to 2025-12-31)	17.8 MB	NOAA public data

Sources: the TLC trip record page and `taxi_zone_lookup.csv` on `d37ci6vzurychx.cloudfront.net`, and the NOAA GHCN daily file on `ncei.noaa.gov`. Everything is downloaded by `src/download_data.py`.

Joins: trips and zones on the pick up zone id, and trips and weather on the pick up date. Both joins are done in `notebooks/02_analysis.ipynb`, after cleaning, on the table of trips per zone per day (not on the 101 million raw rows).

Why Spark: the trips are about 108.7 million rows, which is too much for pandas on a laptop (the files are 1.86 GB compressed and a lot bigger in memory). Spark reads the data in parts and runs the joins and aggregations in parallel on all cores.

3. Pipeline



Pipeline from raw data to results

1. **Download** (`src/download_data.py`): gets the 30 trip files, the zone csv and the weather csv. It retries a failed file 4 times and skips files that are already there.
2. **Ingest** (`src/ingest.py`, `src/schema.py`): reads the raw files with one explicit schema and writes typed Parquet tables. The trips are read month by month. The weather is cut to our 915 days and 5 columns.
3. **Clean** (`notebooks/01_clean.ipynb`): removes bad trips (rules in section 4), converts the weather to mm and celsius, and saves the row counts. It writes `trips_clean` and `weather_clean`.
4. **Analysis and plots** (`notebooks/02_analysis.ipynb`, Spark SQL and DataFrame API): labels every day (rain, snow, cold, hot), counts trips per zone per day, joins with weather and zone names (this is where both joins happen), compares every weather type to a normal day, and draws 12 charts.

The download and the ingest are scripts, because they run once and have nothing to look at. The cleaning and the analysis are notebooks, so every step shows its result right under the code. Everything runs in Docker with the image from our lab (Jupyter with PySpark 3.5.0), and the README has the exact commands.

4. Data cleaning

What we found in the trips (108,708,928 rows):

- 115 trips with a pick up date outside July 2023 to December 2025 (years 2002 to 2009, and a few from June 2023 and June 2026).
- 5,301 trips where the drop off is before the pick up, 562,648 with zero duration and 1.2 million shorter than one minute. 59,725 last more than 3 hours.
- 3.79 million negative fares (credits, voids and adjustments, not normal paid trips: 57% flex fare, 27% dispute, 10% cash, 6% no charge) and 44,959 zero fares. 1,710 fares are above 500 dollars.
- 2.68 million trips with distance 0, and 5,110 with more than 100 miles.
- 412,688 trips start and 968,604 end in the zones “Unknown” (264) or “Outside of NYC” (265).
- 16.5 million trips (15%) have no passenger count and a few other empty fields. All of them are payment type 0 (flex fare).
- Only 7 exact duplicate rows. No zone id outside 1 to 265.

What we did. We started with 108,708,928 trips and applied these rules in this order, so each rule only counts the rows that passed the ones before it. Removing trips with a pick up date outside July 2023 to December 2025 took out 115 rows. Keeping only trip times from 1 to 180 minutes removed 1,875,936 rows. Keeping fares above 0 and at most 500 dollars removed 3,622,198 rows. Keeping distances above 0 and at most 100 miles removed 1,593,787 rows. Removing unknown pick up zones (264 and 265) took out 329,583 rows, and removing unknown drop off zones (264) took out 198,559 rows. Finally, dropping exact duplicates removed 4 rows. In total 7,620,182 trips were removed, leaving 101,088,746 trips (93.0%). We also keep only the 11 columns we need and add `pickup_date` and `duration_min`.

Weather (915 rows): we checked missing values, duplicate days and impossible values (negative rain or snow, temperatures out of range, min above max). Nothing had to be removed, so 915 days stay 915 days. The only change is the units: NOAA gives rain and temperature in tenths, we convert to mm and celsius.

5. Important design decisions

- **Explicit schema, one for all months.** The 2025 files have one extra column (`cbd_congestion_fee`, the new congestion fee). It is in the schema for all months and is null for 2023 and 2024.
- **Month by month, only needed columns.** Trips are ingested one month at a time, and the cleaned table keeps 11 of the 20 columns. This keeps memory low on a laptop. The analysis first shrinks the trips to one row per zone per day (about 209,000 rows) and caches it, so Spark does not scan 101 million rows again for every query.
- **Cleaning limits.** The limits are wide on purpose: for fares above 0 the 99.9th percentile is about 150 dollars and for distances it is about 30 miles, so 500 dollars and 100 miles only remove clear errors. Negative fares are credits and adjustments, not normal paid trips, so they go. Zero distance trips are removed because the proposal says so and a trip without distance is not a normal trip.
- **We keep missing passenger counts.** 13 million cleaned trips have no passenger count. We never use that column, so deleting 13% of the trips for it would hurt the analysis for nothing.
- **Zones.** The analysis is per pick up zone, so pick ups from “Unknown” and “Outside of NYC” are removed. We keep drop offs at “Outside of NYC” because those trips are real and part 3 can use them.
- **Weather types** (from the daily data): rain is 1 mm or more (257 days), snow is any snow (22 days), cold is a daily max under 5 C (94 days), hot is 25 C or more (308 days), everything else is mild. The normal day for every comparison is: dry, no snow, mild.
- **Trips per day, not total trips.** There are 658 dry days but only 22 snow days, so totals would not be comparable.
- **Only busy zones.** Zones with less than 100 trips per day on normal days are left out, because a few trips more or less gives huge percentages. 66 of the 263 zones are left, and they cover 96.5% of the cleaned trips.
- **Scripts and notebooks.** Download and ingest are scripts (run once, no results to look at). Cleaning and analysis are notebooks, so every step shows its result.
- **Docker from the lab, shared project folder.** Anyone can run the pipeline with the README commands, no Java or Spark install needed. The project folder is shared with the container, so the data stays on the computer and is not lost when the container stops.

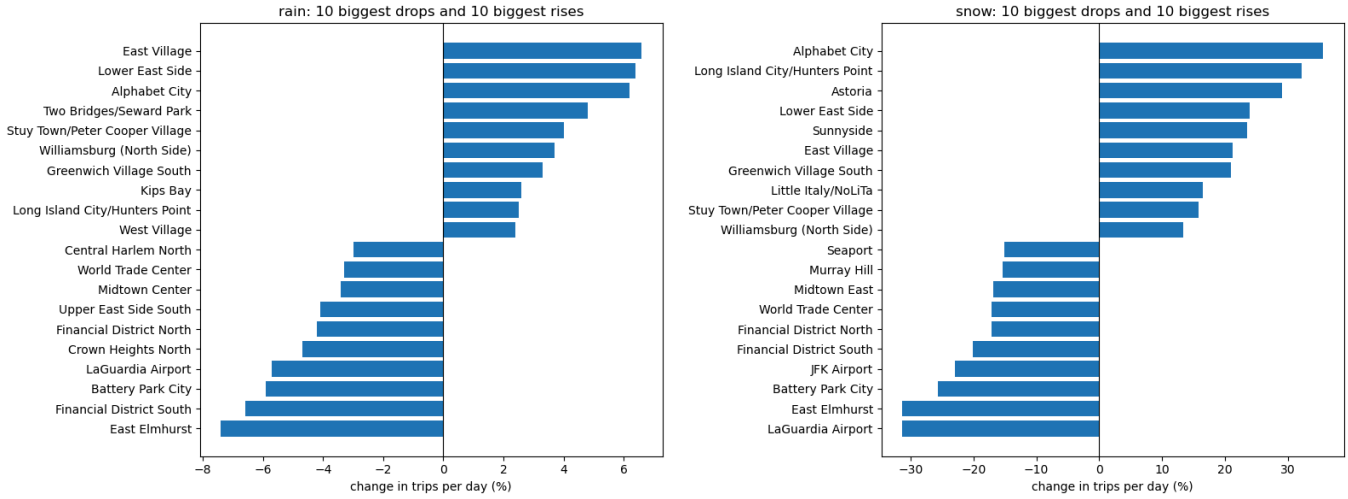
6. Answer to the sub-question

Weather changes trips, trip time and fares a little in the city as a whole, but a lot in some zones. The table shows the whole city (all zones). Every cell is the weather day first and the normal day second:

Weather	Days	Compared with	Trips per day	Trip time (min)	Average fare (dollars)
rain	257	dry days (658)	109,789 vs 110,749 (-0.9%)	17.05 vs 17.16 (-0.6%)	19.83 vs 19.94 (-0.6%)
snow	22	days without snow (893)	104,858 vs 110,618 (-5.2%)	14.72 vs 17.18 (-14.3%)	18.37 vs 19.95 (-7.9%)
cold (max under 5 C)	94	mild days (513)	109,928 vs 113,144 (-2.8%)	16.22 vs 17.25 (-6.0%)	19.25 vs 19.83 (-2.9%)
hot (max 25 C or more)	308	mild days (513)	106,210 vs 113,144 (-6.1%)	17.19 vs 17.25 (-0.3%)	20.25 vs 19.83 (+2.1%)

Per zone (66 busy zones):

- **Rain has a small effect.** Trips per day change between -7.4% and +6.6%, and the median zone loses 0.7%. Trip time and fare hardly move (median -0.3% and 0.0%). Airports and the Financial District lose a bit (East Elmhurst -7.4%, Financial District South -6.6%), and East Village (+6.6%) and Lower East Side (+6.4%) gain a bit.
- **Snow has the biggest effect.** LaGuardia Airport and East Elmhurst lose 31% of their trips, Battery Park City 26% and JFK 23%. Queens as a whole loses 20% of its trips, Manhattan 3.8%. Some zones gain: Alphabet City +36%, Long Island City +32%, Astoria +29%. Trip time is shorter in all 66 zones (median -11.5%) and fares are lower in 62 zones (median -4.9%), so the trips that still happen are shorter.
- **Cold:** 45 of 66 zones lose trips (median -2.7%). East Elmhurst -16.5%, Seaport -13.7% and the airports about -11%. Astoria gains 10.3%. Trip time is lower in 64 zones.
- **Hot:** 47 of 66 zones lose trips (median -3.8%), mostly in Manhattan (-7.5% as a whole). Upper East Side North -19.0% and Upper West Side North -16.1%. Williamsburg gains 18.7% and Queens gains 3.9%. Trip time and fares barely change.



Percent change in trips per day for the 10 biggest drops and rises, on rainy days (left) and snow days (right).

Limits. There are only 22 snow days, so the snow results are noisy. The weather comes from one station (Central Park) for all zones. Hot days are summer days and cold days are winter days, so temperature is mixed with the season (holidays, tourists). We compare averages, so this does not prove that the weather causes the change. Whether bad weather changes demand more than past data predicts is the job of part 2.